

DualBand: Supporting On-Body Touch Input Using Two Wristbands with RF Sensing

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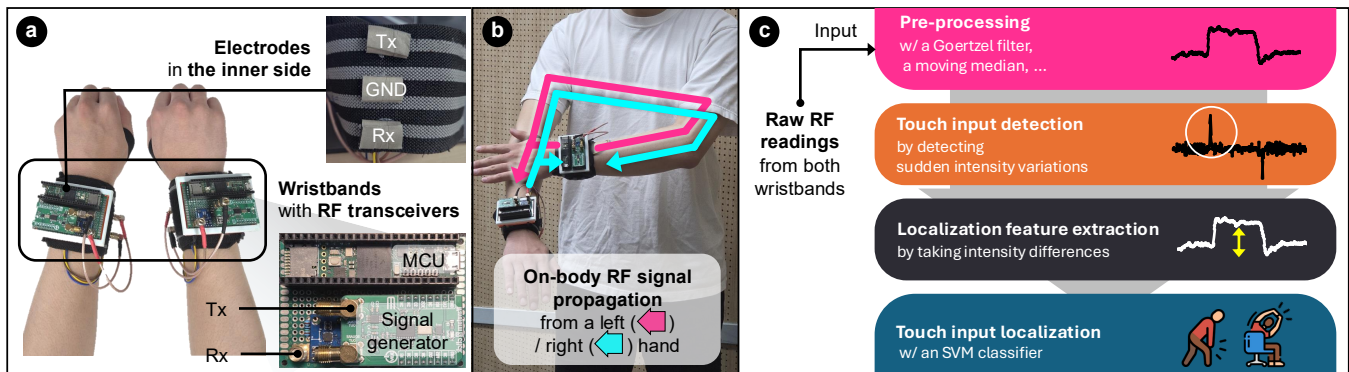


Figure 1: Overview of DualBand. (a) DualBand supports on-body touch interaction using two wrist-worn devices. (b) RF signals propagate through the body. (c) DualBand analyzes how RF signals propagate through the body to detect touch inputs and estimate their locations.

Abstract

The commercialization of virtual and augmented reality technologies is creating a growing demand for novel input methods that enable intuitive and accessible interaction. To this end, we present DualBand, a wearable on-body touch input system capable of tracking touch locations across diverse body regions. DualBand employs two wrist-worn devices that act as transceivers, injecting low-power RF signals into the body and capturing the resulting received signals. It then detects and localizes touch input by extracting their intensity variations. Experimental results across seven body parts demonstrates an overall localization accuracy of 81.5%. Future work will investigate improving localization performance by combining multimodal sensors.

CCS Concepts

• Human-centered computing → Interaction devices.

Keywords

On-body touch interaction, Wrist-worn device, On-body RF signal propagation

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1 Introduction

Recent advances in virtual and augmented reality technologies have introduced new human-computer interaction paradigms, creating a growing need for novel input methods. In light of this trend, on-body input methods, which uses a human body as a touch input surface, have been widely studied [1–3, 5–12]. These techniques not only support intuitive interaction but also enable precise manipulation even in eye-free situations based on human proprioception.

In particular, as a promising solution to enable on-body touch interaction, many works [5, 9, 10, 12] have leveraged RF (Radio Frequency) signals propagating through the body. Specifically, most of them propose wristband- or ring-like wearable devices that inject low-power RF signals into the human body and capture the body-propagated signals to localize touch positions on the body. However, they still have limitations that hinder their wide adoption in diverse real-world applications; they allow on-skin touch interaction only

over small regions of the body (e.g., arms and hands) or support only limited types of touch gestures (e.g., simple finger pinches).

To address these limitations, we introduce DualBand, a novel on-body touch input system capable of tracking unimanual touch locations across various body parts. The key idea behind DualBand is to leverage the propagation patterns of RF signals traveling through the body, which vary depending on the touch location.

2 DualBand Design and Implementation

As shown in Figure 1, DualBand employs two wearable devices worn on each wrist to determine the timing and location of touch events on the human body through the following steps. First, the devices act as transceivers. They generate single-tone signals with a signal generator and injects them into the body at an output current of 4 mA, which remains regulatory exposure limits [4]. In particular, to differentiate the signals from each device, they transmit signals at different frequencies of 10.04 MHz and 10.21 MHz, respectively. The body-propagated signals are received by both devices, converted into digital signals using their built-in ADCs at a sampling rate of 500 kHz with 8-bit resolution, and sent to a server for further analysis to identify when and where touch events occur. Note that we design easy-to-wear and adjustable wearables by combining the transceiver devices with flexible sports wrist straps (see Figure 1(a)). The inner side of each strap consists of three electrodes, referred to as Tx, GND, and Rx. Each electrode connects to the corresponding port on the transceiver hardware via an SMA cable and makes contact with the wrist skin. In this setup, DualBand transmits and receives RF signals through the Tx and Rx electrodes, while the GND electrode placed between Tx and Rx helps enhance the clarity of the received signals.

As illustrated in Figure 1(c), After receiving the RF signal readings from each wristband, the server detects on-body touch events by identifying the time points at which sudden variations in the received signals occur. For each transmitter–receiver pair, we compute the received signal strength using a Goertzel filter and calculate the moving median over a 50 ms sliding window. A touch event is detected if the moving median from both wristbands increases by more than 3 relative to the preceding window. Once a touch event is detected, DualBand extracts localization features by computing the difference in received signal intensity before and after the event for each transmitter–receiver pair. In other words, it leverages the effect of propagation path variations caused by the input on the received intensity. Finally, DualBand identifies the touch position by using the extracted features as input to a Support Vector Machine (SVM) classifier trained on pre-collected labeled data.

3 Evaluation

To evaluate the feasibility of enabling on-body touch interaction with DualBand, we conducted experiments with a single participant under real-world conditions. As shown in Figure 2, the participant was instructed to touch each input position using their hand while wearing the DualBand wrist-worn devices and standing freely. Considering all possible combinations of hand and input position, the participant performed eight distinct on-body touch inputs. This process was repeated five times for each input. Using the collected

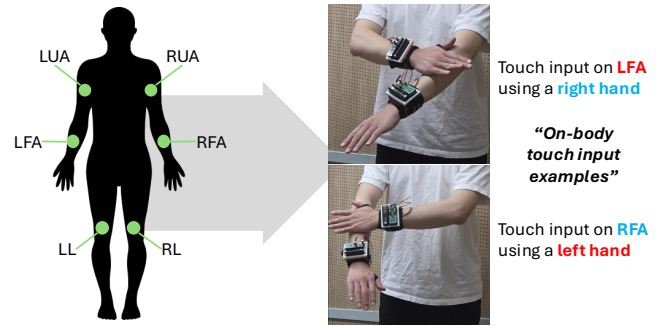


Figure 2: On-body touch input positions used for experiments and interaction examples. Positions include the left forearm (LFA), left upper arm (LUA), right forearm (RFA), right upper arm (RUA), left leg (LL) and right leg (RL).

[illegible]

Figure 3: Performance of DualBand in identifying on-body touch inputs.

dataset, we trained an SVM classifier with an RBF kernel and measured touch localization accuracy via 5-fold cross-validation.

As illustrated in Figure 3, DualBand achieves a reasonable localization accuracy of 81.5% on average. Notably, the localization performance degrades significantly when the user touches the legs. More specifically, DualBand struggles to differentiate touch inputs made on the legs using the same hand. This is because the signal propagation paths newly formed between the two wristbands during leg touches are both relatively long and similar in length, which results in minimal changes in received signal intensity and reduced distinctiveness of the features used for localization. However, excluding touch inputs on the legs, the accuracy increases to 98.3% on average, demonstrating the feasibility of supporting on-body touch interaction across the upper body.

4 Conclusion

In this paper, we presented DualBand, a wearable on-body touch input system that leverages propagation patterns of low-power RF signals to detect and localize touch events across diverse body regions using only two wrist-worn devices. Our evaluation demonstrated localization accuracy of up to 81.5% across seven body parts, highlighting the feasibility of supporting intuitive and accessible on-body touch interaction. However, there is still room for further improvement. Therefore, we plan to explore the collaborative use of multimodal sensors for more precise and robust on-body touch localization. Additionally, future work will involve validating DualBand with a larger pool of users and assessing its robustness under diverse conditions, including variations in wearing posture and different clothing materials, to evaluate real-world applicability.

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